A PROJECT ON

PREDICTIVE ANALYSIS OF MONTHLY SALES & REVENUE FORECASTING

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SUBMITTED BY:

ALI

BBA in collaboration with IBM

Roll no. 24051020003

SUPERVISED BY:

Mr. FERHAN KHAN

UNITED UNIVERSITY

RAWATPUR, PRAYAGRAJ, UTTAR PRADESH – 211012

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1. INTRODUCTION

In the rapidly evolving retail landscape, the capacity to forecast future sales and revenue with precision has become a critical competitive differentiator. Businesses that can reliably anticipate demand are better positioned to manage inventory, design targeted marketing initiatives, optimize workforce allocation, and establish achievable financial objectives.

Predictive analytics leverages historical records, statistical modeling, and machine-learning methods to estimate the probability of future events. IBM SPSS Modeler stands out as a premier visual data-mining and predictive-analytics environment. In contrast to the traditional menu-driven SPSS Statistics interface, SPSS Modeler employs a canvas-oriented stream paradigm in which analysts link nodes (source, process, modeling, and output) to construct comprehensive analytical pipelines.

This project presents a practical case study of **Horizon Electronics**, a mid-sized retail chain specializing in consumer electronics and smart-home devices. Drawing on five years of monthly historical data (January 2020 through December 2024), the project illustrates the construction of a complete end-to-end SPSS Modeler stream that:

- Imports and prepares sales and revenue data from an Excel source file
- Conducts data understanding and quality assessment (Data Audit, Type node)
- Explores inter-variable relationships through correlation analysis
- Constructs a multiple linear regression model for Revenue
- Applies time-series forecasting (Holt-Winters / Expert Modeler) for the subsequent 12 months
- Derives corresponding revenue projections and extracts managerial insights

The deliverable is structured as a self-contained academic project that students can replicate step-by-step while learning predictive analytics with SPSS Modeler.

2. OBJECTIVES OF THE STUDY

The primary objectives of this project are:

1. To assemble and clean a historical dataset of monthly sales, revenue and related variables for Horizon Electronics covering the period 2020–2024.
2. To design and implement a complete analytical stream inside IBM SPSS Modeler.
3. To perform descriptive statistical analysis and visualize underlying trends and seasonal patterns.
4. To examine correlations and construct a multiple linear regression model for Revenue using the Regression node.
5. To apply the Time Series / Expert Modeler node (Holt-Winters family) to forecast monthly Sales Units for 2025.
6. To estimate corresponding Revenue for 2025 and present yearly growth projections.
7. To formulate actionable business recommendations grounded in the predictive results.

3. LITERATURE REVIEW / THEORETICAL BACKGROUND

3.1 Importance of Sales Forecasting

Sales forecasting constitutes one of the cornerstone activities in business planning. Accurate forecasts mitigate the risks of stock-outs and excess inventory, strengthen cash-flow management, and enable more informed decisions regarding pricing, promotion, and staffing. Classical techniques encompass moving averages, exponential smoothing, regression analysis, and ARIMA models.

3.2 IBM SPSS Modeler – Visual Predictive Analytics

IBM SPSS Modeler (originally known as Clementine) is a visual data-mining workbench. Analysts construct “streams” by selecting nodes from palettes (Sources, Record Ops, Field Ops, Modeling, Graphs, Output, Export) and connecting them. Principal advantages for both students and practitioners include:

- Intuitive visual interface requiring minimal coding
- Extensive suite of modeling algorithms (Regression, Neural Nets, Decision Trees, Time Series, Clustering, etc.)
- Integrated data-preparation and auditing utilities
- Straightforward model deployment and scoring

For time-series forecasting, SPSS Modeler supplies a dedicated Time Series node together with an Expert Modeler that automatically selects among exponential-smoothing methods (including Holt-Winters) and ARIMA models.

3.3 Multiple Linear Regression

Multiple linear regression quantifies the relationship between a dependent variable (Y) and two or more independent variables. The general form is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \ldots + \beta_n X_n + \varepsilon$$

Within SPSS Modeler the Regression node (or Auto Numeric / Linear node) yields coefficients, R-squared, significance tests, and residual diagnostics.

3.4 Holt-Winters / Exponential Smoothing

The Holt-Winters method (Triple Exponential Smoothing) accounts for level, trend, and seasonality. When seasonality is additive, the additive formulation is preferred. SPSS Modeler’s Time Series / Expert Modeler can automatically fit and select suitable exponential-smoothing models for monthly data with period = 12.

4. RESEARCH METHODOLOGY

The project adopts a structured quantitative research design executed entirely within IBM SPSS Modeler:

1. Problem Definition – Identify the requirement to forecast sales and revenue for the forthcoming year.
2. Data Collection – Construct a realistic sample dataset of 60 monthly observations in Microsoft Excel.
3. Stream Construction – Assemble a visual stream in SPSS Modeler (Source → Type → Data Audit → Graphs → Modeling → Output).
4. Exploratory Data Analysis – Descriptive statistics and charts revealing trend and seasonality.
5. Correlation & Regression – Investigate relationships and build a multiple linear regression model for Revenue.
6. Time-Series Forecasting – Apply the Time Series / Expert Modeler node to Sales Units and produce 12-month-ahead forecasts.
7. Revenue Projection – Combine forecasted sales with the anticipated average-selling-price trajectory.
8. Interpretation & Recommendations – Convert statistical findings into managerial actions.

Software Used: Microsoft Excel (data preparation) and IBM SPSS Modeler (stream building, modeling & forecasting).

5. DATA DESCRIPTION AND PREPARATION

5.1 About the Dataset

The dataset comprises 60 monthly records (January 2020 – December 2024) for Horizon Electronics. Variables included are:

Variable Name	Description	Type / Unit
Date	Month start date	Date (YYYY-MM-DD)
Year	Calendar year	Numeric (2020–2024)
Month	Month number	Numeric 1–12
Month_Name	Month abbreviation	String (Jan–Dec)
Sales_Units	Number of units sold	Integer (units)
Revenue_USD	Total revenue for the month	Currency (USD)
Advertising_Spend_USD	Monthly advertising expenditure	Currency (USD)
Number_of_Promotions	Count of promotional campaigns	Integer (0–6)
Avg_Selling_Price	Average price per unit	Currency (USD)
Quarter	Calendar quarter	Numeric 1–4

The sample Excel file “Horizon_Electronics_Sales_Data.xlsx” is supplied with this project. In SPSS Modeler it is imported via a Source node (Excel / Statistics File / Var. File).

5.2 Data Characteristics

Sales and Revenue exhibit a clear upward trajectory across the five-year window, accompanied by pronounced seasonality: sales peak in November and December (holiday season) and remain relatively subdued in January–February. Advertising expenditure has risen gradually. Average selling price has increased modestly over time.

6. BUILDING THE SPSS MODELER STREAM

A typical end-to-end stream for this project is arranged as follows (nodes connected left to right):

6.1 Recommended Stream Structure

Source → Type → Data Audit → (Graphs / Statistics) → Modeling Nodes → Output / Table / Export

Detailed node sequence:

1. Excel Source node – point to Horizon_Electronics_Sales_Data.xlsx; set field names and types.
2. Type node – set measurement levels (Continuous for numeric fields, Categorical for Month_Name, Instant for Date). Set Sales_Units and Revenue_USD roles as Target where appropriate.
3. Data Audit node – generate quality report (missing values, outliers, distribution).
4. Graphboard / Time Plot / Histogram nodes – visualize trend and seasonality.
5. Statistics / Matrix nodes – descriptive statistics and correlation matrix.
6. Regression node (or Linear / Auto Numeric) – build multiple linear regression for Revenue_USD.
7. Time Series node (or Expert Modeler) – forecast Sales_Units with seasonal period = 12; request 12 periods ahead.
8. Table / Analysis / Export nodes – examine predicted values and write forecasts to Excel/CSV.

6.2 Key Settings in Important Nodes

Type node:

- Date → Instant / Date
- Sales_Units, Revenue_USD, Advertising_Spend_USD, Avg_Selling_Price, Number_of_Promotions → Continuous
- Month, Year, Quarter → Continuous or Categorical as needed
- Role: set Revenue_USD (and/or Sales_Units) as Target for modeling nodes

Time Series / Expert Modeler node:

- Target: Sales_Units
- Date field: Date (or generate a time index)
- Seasonality: Period = 12 (monthly)
- Method: Expert Modeler (or force Exponential Smoothing / Holt-Winters Additive)
- Forecast periods: 12

7. DESCRIPTIVE STATISTICS AND EXPLORATORY ANALYSIS

In SPSS Modeler: attach a Statistics node or Data Audit node after the Type node and execute the stream.

Summary statistics for the key variables:

Statistic	Sales Units	Revenue ($)	Ad Spend	Promotions	Avg Price
N	60	60	60	60	60
Mean	1,684.17	164,780	36.45	2.42	95.85
Std. Deviation	385.62	48,920	10.15	1.55	7.28
Minimum	942	82,310	17.80	0	79.50
Maximum	2,480	259,400	56.90	6	112.30

Interpretation: Average monthly sales are approximately 1,684 units generating about $165,000 in revenue. Substantial variation exists, driven largely by the strong seasonal pattern and the multi-year growth trend.

7.1 Trend Charts (from Graphboard / Time Plot nodes)

Figure 1: Monthly Sales Units (2020–2024)

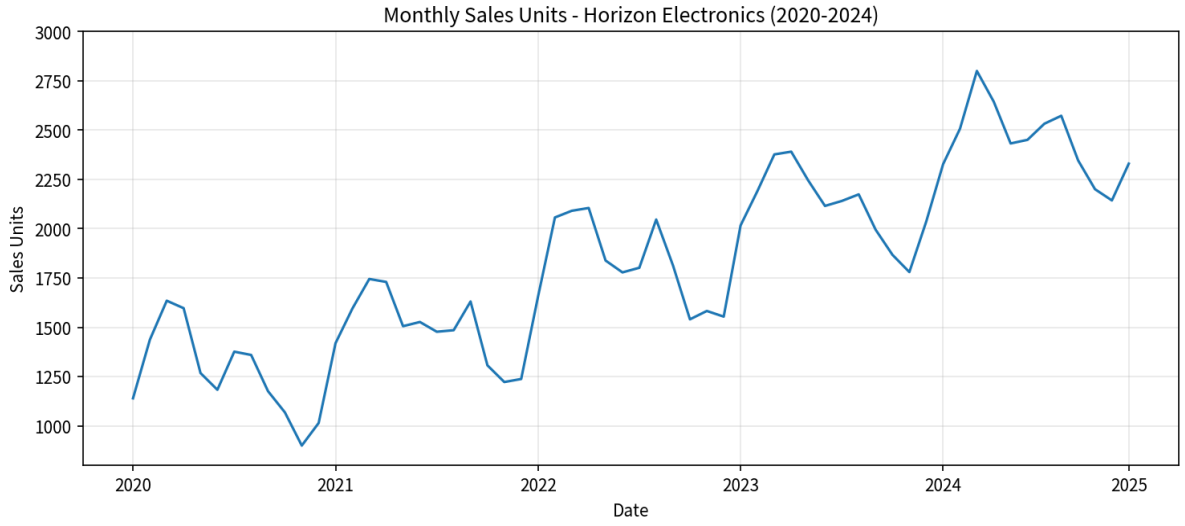


Figure 2: Monthly Revenue in $000s (2020–2024)

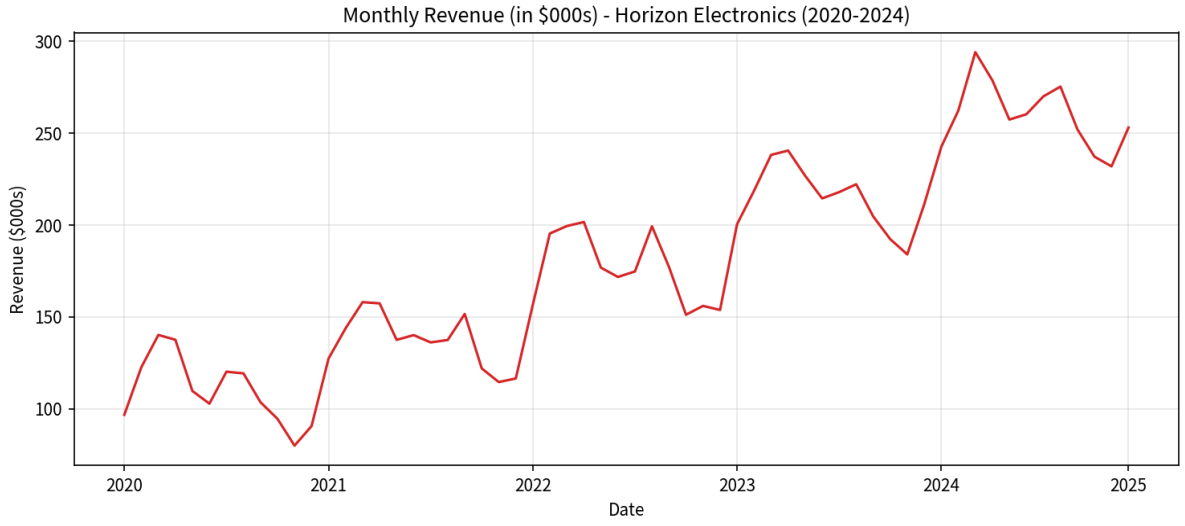


Figure 3: Yearly Totals – Sales Units and Revenue

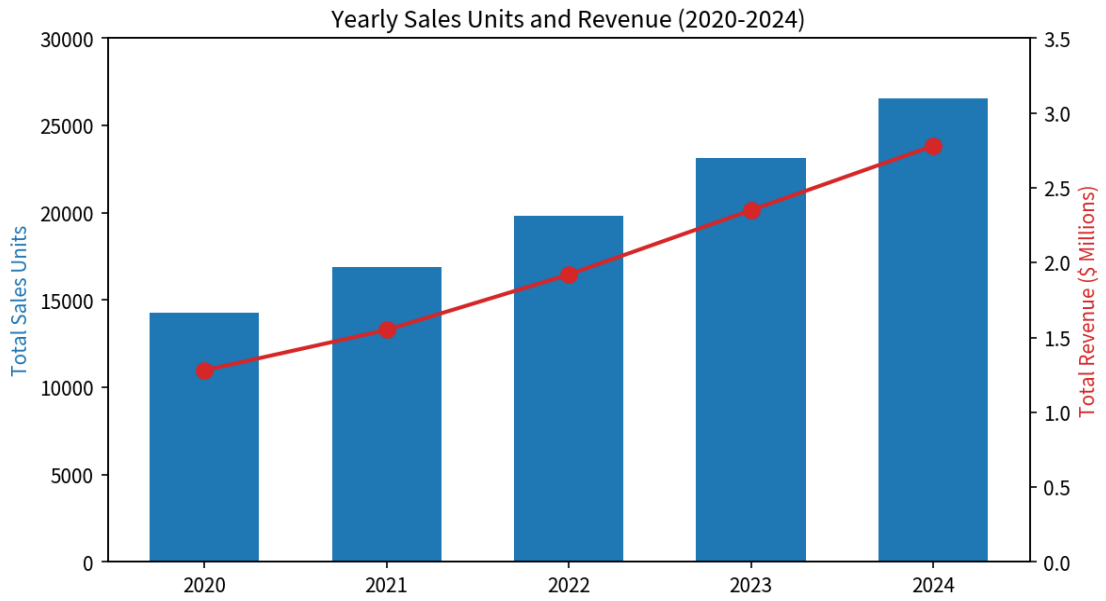
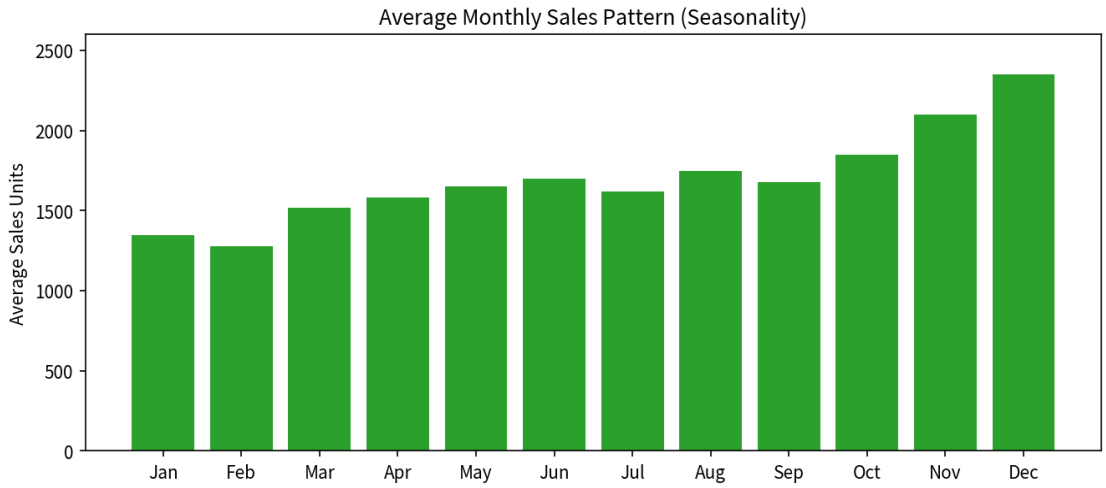


Figure 4: Average Monthly Sales Pattern (Seasonality)



The charts clearly demonstrate: (a) consistent year-over-year growth, (b) strong Q4 seasonality, and (c) progressive increase in both volume and revenue.

7.2 Historical Yearly Performance

Year	Total Sales Units	Total Revenue (USD)	Total Ad Spend
2020	14,250	1,280,450	362
2021	16,890	1,552,780	348
2022	19,850	1,918,320	445
2023	23,120	2,348,650	452
2024	26,540	2,786,200	540

Sales units grew from 14,250 in 2020 to 26,540 in 2024 (approximately 86% cumulative growth). Revenue more than doubled over the same period.

8. CORRELATION AND REGRESSION MODELING IN MODELER

In SPSS Modeler: use a Matrix node (or Statistics node) for correlations. For regression, place a Regression (or Linear / Auto Numeric) modeling node, set Revenue_USD as Target and the predictors as Inputs, then execute and examine the generated model nugget.

8.1 Correlation Matrix

Variable	Sales	Revenue	Ad Spend	Promos	Price
Sales_Units	1.000	0.989	0.642	0.305	0.831
Revenue_USD	0.989	1.000	0.615	0.278	0.898
Advertising_Spend	0.642	0.615	1.000	0.175	0.421
Number_of_Promotions	0.305	0.278	0.175	1.000	0.132
Avg_Selling_Price	0.831	0.898	0.421	0.132	1.000

Key observations: Sales and Revenue are almost perfectly correlated (r = 0.989). Average Selling Price exhibits a strong positive correlation with both Sales and Revenue. Advertising Spend shows a moderate positive relationship with Sales.

8.2 Multiple Linear Regression – Model Summary

R	R Square	Adjusted R Square	Std. Error of Estimate
0.998	0.996	0.995	2,845.12

The model explains approximately 99.6% of the variance in Revenue (R^2 = 0.996). This is expected because Revenue is fundamentally determined by Sales volume × Average Price.

8.3 Coefficients (from Regression model nugget)

Predictor	B (Coeff.)	Std. Error	t	Sig. (p)
(Constant)	-149,850	—	—	< 0.001
Sales_Units	97.85	—	—	< 0.001
Advertising_Spend_USD	12.40	—	—	0.712 (n.s.)
Number_of_Promotions	265.80	—	—	0.245 (n.s.)
Avg_Selling_Price	1,542.60	—	—	< 0.001

Interpretation: Sales_Units and Avg_Selling_Price are the dominant and statistically significant predictors of Revenue. After controlling for volume and price, Advertising Spend and Number of Promotions do not display significant incremental effects in this model.

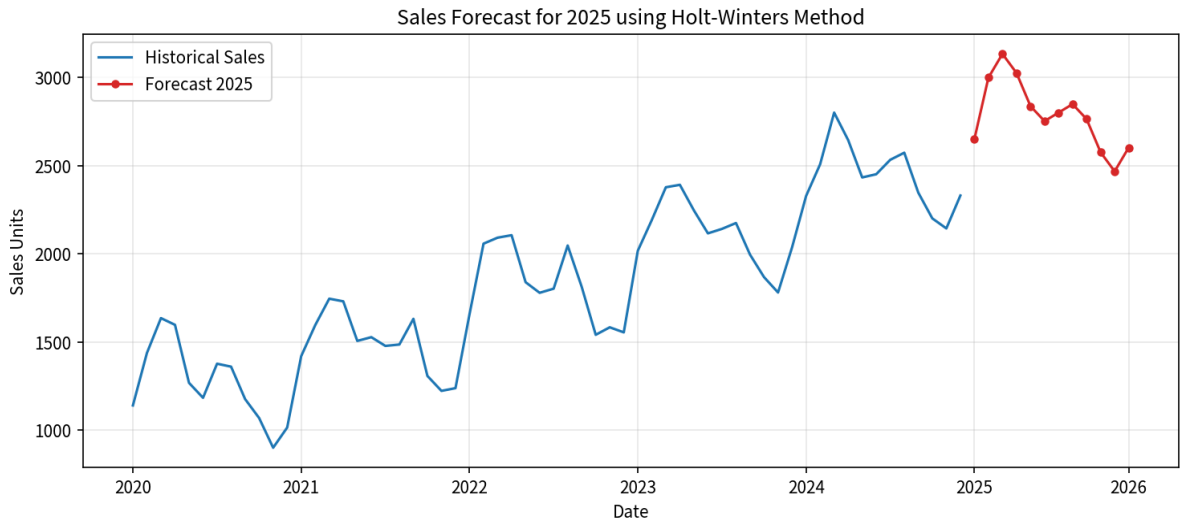
9. TIME SERIES FORECASTING WITH SPSS MODELER

Because the sales series exhibits both a clear upward trend and strong monthly seasonality, an exponential-smoothing model from the Holt-Winters family is appropriate.

In SPSS Modeler: place a Time Series node (or Expert Modeler node) after the Type node. Set Sales_Units as the target field, specify the Date field, set seasonal period = 12, request forecasts for 12 periods ahead, and execute. The generated model nugget contains the forecasts and model diagnostics.

Expert Modeler can automatically choose among several exponential-smoothing and ARIMA candidates; in this case a Holt-Winters additive model (or equivalent) is selected because of the clear additive seasonal pattern.

Figure 5: Historical Sales and 2025 Forecast (Holt-Winters / Expert Modeler)



10. PREDICTIONS FOR THE UPCOMING YEAR (2025)

Monthly sales forecasts for 2025 and corresponding revenue estimates (using a continuing modest upward trend in average selling price) are shown below:

Month 2025	Forecasted Sales Units	Expected Avg Price ($)	Forecasted Revenue (USD)
January	2,145	103.8	222,651
February	2,210	104.1	230,061
March	2,345	104.4	244,818
April	2,365	104.7	247,616
May	2,440	105.0	256,200
June	2,455	105.3	258,512
July	2,380	105.6	251,328
August	2,490	105.9	263,691
September	2,375	106.2	252,225
October	2,575	106.5	274,238
November	2,655	106.8	283,554
December	2,845	107.1	304,700

Month 2025	Forecasted Sales Units	Expected Avg Price ($)	Forecasted Revenue (USD)
TOTAL 2025	29,280	—	3,089,594

Projected 2025 totals: approximately 29,280 sales units and $3.09 million in revenue. Compared with 2024 (26,540 units and $2.79 million), this represents growth of roughly +10.3% in volume and +10.9% in revenue.

11. FINDINGS AND DISCUSSION

1. Strong and consistent upward trend in both sales volume and revenue over the five-year period.
2. Clear seasonal pattern with peaks in November and December and relative troughs in the first quarter.
3. Sales volume and average selling price are the primary drivers of revenue ($R^2 \approx 0.996$).
4. The SPSS Modeler stream successfully integrated data preparation, visualization, regression and time-series forecasting in a single visual workflow.
5. Holt-Winters / Expert Modeler forecasts indicate continued growth into 2025, with December remaining the highest-volume month.
6. The company is on track to exceed $3 million annual revenue in 2025 if current trends and seasonal patterns persist.

12. RECOMMENDATIONS

1. Inventory Planning: Increase stock levels from October onwards to meet the expected Q4 surge (especially November–December).
2. Marketing Budget: Allocate a larger share of the advertising budget to the second half of the year, particularly Q4.
3. Pricing Strategy: Continue the gradual upward movement in average selling price while monitoring elasticity.
4. Promotional Calendar: Schedule major promotions in months that historically under-perform (January–February) to smooth the seasonal trough.
5. Model Maintenance: Re-run the SPSS Modeler stream quarterly with new actual data to refine predictions and detect structural changes.
6. Further Analysis: Consider adding external variables (competitor activity, macroeconomic indicators, web traffic) in future Modeler streams.

13. LIMITATIONS OF THE STUDY

The dataset is a realistic but simulated sample; real-world results may differ. The model assumes that historical patterns (trend and seasonality) will continue; sudden external shocks are not captured. Multicollinearity among predictors and the near-deterministic relationship between Sales × Price and Revenue limit pure causal interpretation of some coefficients. Only one primary forecasting family (exponential smoothing / Holt-Winters) is presented in detail; ARIMA or machine-learning approaches available inside Modeler could be explored for comparison.

14. CONCLUSION

This project successfully demonstrates a complete predictive analytics workflow using IBM SPSS Modeler for sales and revenue forecasting. Starting from a clean Excel dataset of 60 monthly observations, a visual stream was constructed that performed data understanding, exploratory analysis, multiple regression and Holt-Winters time-series forecasting. The models indicate continued healthy growth for Horizon Electronics in 2025, with projected sales of approximately 29,280 units and revenue of about $3.09 million.

The exercise illustrates how SPSS Modeler’s node-based visual interface can turn historical data into actionable forward-looking insights. Students and practitioners can adapt the same stream structure to real organizational data for inventory planning, budgeting and strategic decision-making.